INCH - POUND
MIL-P-60009B (MI)
24 September 1990
SUPERSEDING
MIL-P-60009A (MI)
1 November 1974

MILITARY SPECIFICATION
PROPELLANT CHARGE ASSEMBLY, ROCKET HIGH EXPLOSIVE, 66 MM, M72A1, M72A2, OR M72A3, WITH COUPLER

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the fabrication of parts, the manufacture of propellant, and the assembly of one type of propellant charge designated as Charge Assembly, Propellant, for use in Rocket, HE, 66 mm, M72A1, M72A2, or M72A3, with Coupler.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340
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SPECIFICATIONS

FEDERAL

QQ-P-416	-	Plating, Cadmium (Electrodeposited)

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-P-217	-	Potassium Perchlorate
MIL-P-14737	-	Propellant, M7
MIL-I-45208	-	Inspection System Requirements
MIL-M-51103	-	Magnesium Oxide, Technical

STANDARDS

FEDERAL

FED-STD-101	-	Packaging Materials, Test Procedures for

MILITARY

MIL-STD-1167	-	Ammunition Data Card
MIL-STD-1168	-	Ammunition Lot Numbering

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY - MI

10048508	-	Charge Assembly, Propellant
10048514	-	Plate, Stud
10048515	-	Stud and Propellant Assembly
10048672	-	Box, Packing, Ammunition Propellant Charge
10048673	-	Carton, Packing, Ammunition
10048777	-	Insert
10048778	-	Plastic Jacketed Insert Assembly

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13012553	-	Charge Assembly, Propellant
13012554	-	Plate, Stud (M72A3)

(Requests for copies of drawings should be made to the contracting officer or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Materials and parts. Materials and parts shall be in accordance with applicable drawings and specifications (see 4.5.2).

3.3 Assembly. The propellant charge assembly shall comply with all requirements specified in drawings 10048508 (M72A2) or 13012553 (M72A3) and with all requirements defined in applicable specifications (see 4.5.2).

3.4 Heat seal of sealed bag. The heat seal of the sealed bag shall comply with acceptance criteria defined in MIL-P-116 (see 4.5.3.6).

3.5 Moisture content of packing material. Samples of the packing material shall comply with moisture content acceptance criteria defined in FED-STD-101, Method 5024 (see 4.6.4).

3.6 Propellant composition. Propellant composition shall be in accordance with MIL-P-14737 for Type II M7 propellant. Only Grade A potassium perchlorate, in accordance with MIL-P-217, may be used. Magnesium oxide shall be in accordance with MIL-M-51103, Type I (see 4.5.3.1).

3.7 Bond strength of stud and propellant assembly. An assembly conditioned at 241° Celsius (C) shall not experience a separation of stud from the grain nor a separation of propellant at less than 85 pounds of force when a longitudinal tensile force is applied as specified (see 4.5.3.2 and 4.6.1).

3.8 Ballistics.

3.8.1 Maximum pressure at 60°C. The maximum pressure developed by the propellant charge at 60°C shall be not greater than 9400 pounds per square inch (psi) when fired in the applicable motor parts assembly and closure (see 4.5.3.3 and 4.6.2).

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3.8.2 Action time at -40°C. The action time at -40°C shall be not greater than 13 milliseconds when measured between the 500 psi points on the pressure time trace (see 4.5.3.4 and 4.6.2).

3.8.3 Velocity at 21.1°C. The velocity of the complete rocket, including propellant and igniter, weighing 1035 ± 1 grams, shall be 478 ± 10 feet per second (fps) when temperature conditioned and fired at 21.1°C (see 4.5.3.5 and 4.6.3).

3.8.4 Velocity at -40°C. The velocity of the complete rocket, including propellant and igniter, weighing 1035 ± 1 grams, shall be 443 ± 16 fps when temperature conditioned and fired at -40°C (see 4.5.3.5 and 4.6.3).

3.8.5 Propellant separation. The propellant grain shall not separate from the stud when fired dynamically in a rocket assembly of appropriate weight at operational temperatures of -40°C to 60°C (see 4.5.3.5 and 4.6.3).

3.9 Workmanship. Parts and assemblies shall be free of chips, dirt, grease, rust and other foreign matter that may impair proper functioning. Cleaning methods used shall not be injurious to any parts, nor shall parts be contaminated by the cleaning agents used (see 4.5.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.5.1, 4.5.2 and 4.5.3.

4.3.1 Submission of product. At the time the completed lot of product is submitted to the government for acceptance, the contractor shall supply the following information, if required by the acquisition document, accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of units of product inspected.

c. Drawing, specification number and date, together with an identification and date of changes.

d. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

e. Number of items in the lot.

f. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.3.2 Contractor quality assurance program. The contractor shall provide and maintain an effective quality assurance program in compliance with MIL-I-45208.

4.3.3 Government verification. Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify, at unscheduled intervals, all quality assurance operations performed by the contractor.

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Verification will be in accordance with MIL-I-45208 and will be performed to the extent necessary to ensure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "*critical*" *be accepted solely on the basis of the contractor's records.*

4.3.4 Inspection equipment. The inspection equipment required to perform the examinations and tests required by 4.5.2 and 4.5.3 of this specification shall be designed and fabricated by the contractor.

4.4 First article inspection.

4.4.1 First article. The first article sample shall be examined 100 percent for all of the characteristics in 4.5.2, and then subjected to the tests specified (see 4.4.1.1 and 4.5.3). Inspection and test equipment shall conform to the requirements of 4.3.4. Rejection of the first article sample shall be in accordance with 4.3.3.

4.4.1.1 Protective plating (see drawing 10048514 [M72A2] or drawing 13012554 [M72A3]. The complete quantity of stud plates in the first article sample shall be subjected to the salt spray test specified in QQ-P-416, applicable to the type and class of plating required by the applicable drawing.

4.4.2 Submission of first article. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.5. The first article sample shall consist of the following:

a. 15 consecutively produced stud plates

b. 15 consecutively produced studs.

The first article sample shall consist of material that has been produced by the contractor or furnished by a supplier and that has been manufactured using the same production processes, procedures and equipment that will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to ensure that it conforms to the requirements of the contract and submit a record of this inspection with the sample including statements of findings/certificates of conformance for materials (see 4.3.1d). A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply so as to significantly affect product uniformity as determined by the Government.

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4.5 Quality conformance inspection. Quality conformance inspections shall be as specified in table I.

TABLE I. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph	Method paragraph
First article	3.1	4.4	4.5.3.1
Materials and parts	3.2	4.5.2	Gage/Visual
Assembly	3.3	4.5.2	Gage/Visual
Heat seal of sealed bag	3.4	4.5.3.6	MIL-P-116
Moisture content of packing material	3.5	4.6.4	FED-STD-101, Method 5024
Propellant composition	3.6	4.5.3.1	MIL-P-14737
Bond strength of stud and propellant assembly	3.7	4.5.3.2	4.6.1
Maximum pressure at 60°C	3.8.1	4.5.3.3	4.6.2
Action time	3.8.2	4.5.3.4	4.6.2
Velocity at 21.1°C	3.8.3	4.5.3.5	4.6.3
Velocity at -40°C	3.8.4	4.5.3.5	4.6.3
Propellant separation	3.8.5	4.5.3.5	4.6.3
Workmanship	3.9	4.5.2	Visual

4.5.1 Quality conformance (product lot acceptance inspection).

4.5.1.1 Lot formation. The term "inspection lot", as used throughout this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specifications or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

4.5.1.2 Data cards. Ammunition data cards shall be provided as specified in the contract or purchase order (see 6.2) in accordance with MIL-STD-1167.

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4.5.2 Examination. The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.5.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Equipment necessary for the performance of the inspection listed shall be in accordance with 4.3.4.

4.5.2.1 Insert (see drawing 10048777 covering a detail of drawing 10048515 and table II.

TABLE II. Classification of defects, insert.

Categories	Defects	Method of Inspection
Critical:	Non defined.	
Major:		
101.	Concentricity and shank with diameter of head	Gage
102.	Height of cutoff tip, max.....	Gage
103.	Diameter of head.....	Gage
104.	Angle of head taper.....	Gage
105.	Serrations missing or incomplete	Visual

4.5.2.2 Assembly, plastic jacketed insert (see drawing 10048778, covering a detail of drawing 10048515 and table III).

TABLE III. Classification of defects, assembly, plastic jacketed insert.

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Diameter over plastic, min.....	Gage
102.	Plastic incomplete.....	Visual

4.5.2.3 Plate, stud (see drawing 10048514, covering a detail of drawing 10048508 [M72A2] or drawing 13012554 covering a detail of drawing 13012553 [M72A3] and table IV.

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TABLE IV. Classification of defects, plate, stud.

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Motor seating surface not parallel to closure seating surface.....	Gage
102.	Diameter of holes, max.....	Gage
103.	Location of holes.....	Gage
104.	Depth of counterbore, min.....	Gage
105.	Plating missing or incomplete	Visual
106.	Plate web thickness, max.....	Gage
107.	Web thickness variation, max.....	Gage
108.	Outside diameter, max.....	Gage

4.5.2.4 Stud and propellant assembly (see drawing 10048515, covering a detail of drawing 10048508 [M72A2] or drawing 13012553 [M72A3] and table V.

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TABLE V. Classification of defects, stud and propellant assembly.

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Total length, max.....	Gage
102.	Cracked grain.....	Visual
103.	Evidence of poor workmanship (see 3.9).....	Visual
104.	Location of radial hole.....	Gage
105.	Diameter of radial hole.....	Gage
106.	Solvent missing.....	Visual

4.5.2.5 Charge assembly, propellant (see drawing 10048508 [M72A2] or drawing 13012553 [M72A3] and table VI).

TABLE VI. Classification of defects, charge assembly, propellant.

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Assembly incomplete and/or improper.....	Visual
102.	Evidence of poor workmanship (see 3.9).....	Visual

4.5.6 Carton, packing, prior to sealing top flaps (see drawing 10048673 and table VII).

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TABLE VII. Classification of defects, carton, packing.

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Charge not banded at end opposite from plate end.....	Visual
102.	Packing loose.....	Visual
103.	Moisture content of chipboard sleeves (see 4.6.4).....	Test
104.	Moisture content of fiberboard filler (see 4.6.4).....	Test
105.	Moisture content of carton (see 4.6.4).....	Test

4.5.2.7 Box, packing, ammunition, propellant charge (see drawing 10048672 and table VIII).

TABLE VIII. Classification of defects, box, packing, ammunition, propellant charge.

Categories	Defects	Method of Inspection
Critical:	None defined.	
Major:		
101.	Box broken or split.....	Visual
102.	Strapping missing or loose.....	Visual

4.5.3 Production.

4.5.3.1 Chemical composition and stability of propellant (see 3.6) - Major defect. A sample of propellant shall be selected from each manufacturer's production lot and tested at a Government approved laboratory in accordance with MIL-P-14737. An additional 5-pound sample, consisting of random selections of specimens weighing 0.5 pounds or less shall be selected from each manufacturer's production lot and packaged in airtight containers for surveillance testing. The containers shall be clearly

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marked to indicate the propellant designation, lot number, manufacturer, date of sampling, contract number and number of pounds in the lot. The containers shall be forwarded to the surveillance testing agency as directed by the contracting activity.

4.5.3.2 Strength of stud and propellant assembly (see 3.7 and 4.6.1); separation of stud from grain and/or separation of propellant - Major defect. A sample of 35 stud and propellant assemblies shall be randomly selected from each lot. The test shall be performed as specified in 4.6.1 using test equipment specified in 4.3.4. The lot shall be rejected if any separation occurs at less than 85 pounds of force.

4.5.3.3 Maximum pressure at 60°C (see 3.8.1) - Critical defect. A sample of 20 propellant charges shall be randomly selected from each lot and tested as specified in 4.6.2, using test equipment specified in 4.3.4. The lot shall be rejected if any individual test result is greater than the maximum pressure specified or if the average maximum pressure of the sample plus the product of 1.82 and the standard deviation of the sample is greater than the maximum pressure specified in 3.8.1. The standard deviation of the sample shall be calculated using the formula specified in 6.5. The action time shall be recorded for information only.

4.5.3.4 Action time at -40°C (see 3.8.2) - Critical defect and maximum pressure at -40°C (for information only). A sample of 20 propellant charges shall be randomly selected from each lot and tested as specified in 4.6.2 using test equipment specified in 4.3.4. The lot shall be rejected if the average action time plus the product of 1.82 and the standard deviation of the sample or any individual test result is greater than the action time specified in 3.8.2. The standard deviation of the sample shall be calculated using the formula specified in 6.5. The maximum pressure shall be recorded for information only.

4.5.3.5 Ballistics (see 3.8.3, 3.8.4 and 3.8.5) muzzle velocity at 21.1°C - Major defect; muzzle velocity at -40°C - Major defect; propellant separation - Major defect. A sample of 55 propellant charges shall be randomly selected from each lot, assembled to service motors, and tested as specified in 4.6.3. Thirty test assemblies shall be conditioned at 21.1°C and 20 assemblies shall be conditioned at -40°C for minimum of 6 hours respectively. The remaining five shall be conditioned at 60°C for not less than 6 hours and tested for information purposes only. The lot of charge assemblies shall be rejected if the average velocity of the sample fired at 21.1°C ± the product of 1.88 and the sample standard deviation fails to conform to the requirements of section 3.8.3 or if the average velocity of the sample fired at -40°C ± the product of 2.01 and the sample standard deviation fails to conform to the requirements of section 3.8.4 or, if in the total sample fired, one or more propellant sticks become separated from the stud in two or more assemblies tested. The occurrence of duds caused by igniter failure shall require a replacement test.

4.5.3.6 Heat seal test of sealed bag (see drawing 10048673) - Major defect. Acceptance criteria and test procedure for this test shall be as specified in MIL-P-116..

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4.6 Test methods and procedures. The test methods and procedures described in each of the following tests represent only one method and one procedure which has been proven in use and is approved by the Government for Government acceptance purposes. Producers and manufacturers have the option of using these or substitute methods and procedures. Whichever option is selected, detailed descriptions must be submitted to the Government for approval prior to use. Government approval will be based on an evaluation and demonstration to determine the effectiveness of all of the tests and test procedures.

4.6.1 Strength of stud and propellant assembly. The stud and propellant assembly shall be conditioned at the required test temperature for not less than 6 hours. Reduce the test specimen length to 0.968 + 0.03 inches, and place a steel pin measuring 0.153 inch in diameter and 2.20 inches in length in the stick to prevent wall collapse due to grip force. Place the stud and propellant assembly in an approved test fixture and pull to destruction within 2 minutes after removal from the conditioning chamber. The test equipment shall be operated to exert an increasing force at the rate of 2.5 inches per minute. The assembly shall be classed defective if it does not meet the applicable requirement.

4.6.2 Static functioning. The propellant charge assembly shall be assembled to the applicable test equipment and conditioned at the required temperature ± 1°C for not less than 6 hours. The test shall be completed not less than 2 minutes nor more than 5 minutes after removal from the conditioning chamber. The test equipment shall be capable of obtaining two pressure-time traces simultaneously. Averages of the two traces shall be used in the final data reduction. If one instrumentation channel fails, the results obtained with the remaining channel shall be considered absolute and adequate. Igniters shall be obtained from an established standard lot.

4.6.3 Ballistics. The propellant charge shall be loaded into a service motor with an igniter obtained from an accepted lot and assembled to the test head. The weight of the complete test round shall be 1035 ± 1 grams. The test round shall be conditioned at the required temperature ± 1°C for not less than 6 hours and the test shall be completed in not less than 2 minutes or more than 5 minutes after removal from the conditioning chamber. The test launcher shall be mounted in a rigid horizontal position. Instrumentation to measure the velocity between 5 and 15 feet from the muzzle shall be capable of measuring to 0.1 fps. Any rounds fired during ballistic testing which fail to achieve a velocity within 20 fps of the average velocity for the sample shall be recovered, opened and examined for evidence of propellant loss from the stud. Observations shall be recorded and reported in accordance with contract requirements.

4.6.4 Moisture content of packing material. Samples of the specified packing material shall be selected at the location and time of actual use of that material in the assembly process and from the same lot of the material used to package the lot of

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propellant charge assemblies. The samples selected shall be isolated from the ambient environment prior to testing for moisture content in accordance with FED-STD-101, Method 5024. A charge assembly lot using packing material which fails to satisfy the moisture content requirements of drawing 10048673 shall be classed defective.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified on drawing 10048673.

5.2 Packing and marking. Packing and marking shall be as specified on drawing 10048672.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The propellant charge assembly covered by this specification is intended for use in a Rocket, HE, 66MM, M72A1, M72A2, or M72A3, with coupler.

6.2. Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. Data cards. Data cards must be prepared for each lot in accordance with MIL-STD-1167 (see 4.5.1.2).

c. Propellant description sheets (SMU Form 1129R) shall be attached to each data card describing the propellant composition, processing details of the propellant lot, igniter lot and motor body assembly lot used in the acceptance tests.

e. Whether a first article sample is required (see 3.1 and 4.4).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first articles. Invitations for bids should provide that the government reserves the right to waive the requirement for samples for first

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article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (keyword) listing.

Ballistics
Plate, stud
Powder assembly
Stud and powder assembly

6.5 Calculation of sample standard deviation.

$$S = \sqrt{\frac{\Sigma (x - \bar{x})^2}{n - 1}}$$

where:

s	=	sample standard deviation
x	=	numerical value for individual observations
$\bar{x}$	=	average value for sample
n	=	sample size

6.6 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this document.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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